

Frames and matrix designs

Nonexistence of a complete set of mutually unbiased bases in dimension six

Difficulty: extreme **Importance:** broadly interesting

Status: Open

Rating rationale: Dimension six is the leading composite-dimension obstruction for mutually unbiased bases; its resolution would affect quantum measurements and structured matrix design.

Do there fail to exist seven matrices $U_1, \dots, U_7 \in \mathbb{C}^{6 \times 6}$ such that

$$U_j^* U_j = I_6 \quad (1 \leq j \leq 7), \quad |(U_j^* U_\ell)_{ab}|^2 = \frac{1}{6} \quad (j \neq \ell, 1 \leq a, b \leq 6)?$$

Each U_j is the matrix of an orthonormal basis; the second condition says the bases are pairwise mutually unbiased. The conjecture is $\text{MUB}(6) < 7$. It is weaker than the often discussed assertion $\text{MUB}(6) = 3$, and no claim about that sharper value is included here.

After fixing one basis as the standard basis, the problem becomes simultaneous feasibility of orthogonality and constant-modulus equations for six complex Hadamard matrices. It connects structured matrix construction, Gram matrix feasibility, and certified numerical search. Mutually unbiased measurements also provide well-conditioned quantum state reconstruction designs.

References

1. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 22. [Paper](#).
2. D. McNulty and S. Weigert, *Mutually Unbiased Bases in Composite Dimensions — A Review*, Quantum 10 (2026), article 2051; arXiv:2410.23997v2, revised March 26, 2026. Sections 1 and 7 state and survey the dimension-six existence problem. [Paper](#).
3. M. Cárdenes Wuttig and J. Tindall, *A Complete Classification of Complex Hadamard Matrices of Order Six*, arXiv:2608.18053 (2026). Sections I and VI explicitly distinguish the claimed Hadamard classification from the still unsettled MUB problem. [Paper](#).

Status check — 2026-09-10

Searched “mutually unbiased bases dimension six solved 2026” and “MUB 6 Hadamard classification”; inspected the current 2026 review and August 2026 classification. The latter explicitly states in its conclusion that its result does not settle mutually unbiased bases in $\mathbb{C}^6$. Numerical nonexistence searches for particular families do not exclude all seven-tuples above.

Audit update (2026-09-10): Rechecked the March 2026 MUB review and August Hadamard-classification paper, and searched for dimension-six resolutions. A classification of individual Hadamard matrices does not establish nonexistence of the required mutually unbiased tuple. This is a bounded literature check, not a proof that no solution exists.